

Lab 1 – Requirements Table

Problem Statement #37: Group Travel Expense Simplification Engine

Actors: Group Traveler, Trip Admin

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Section: E

Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a Trip Admin to create a trip and add Group Travelers as participants, each identified by name and a base currency.	High	Pass: A newly created trip shows all added travelers with correct base currencies in the participant list; Fail: any added traveler is missing or has an incorrect currency.	Establishes the foundational trip/participant structure that every other requirement depends on.
FR-002	Functional	The system shall allow a Group Traveler to log an expense with an amount, payer, currency, date, category, and a list of participants who share the cost (equally or by custom split).	High	Pass: A logged expense appears in the trip ledger with the exact amount, payer, and split entered; Fail: the recorded split total does not equal the entered expense amount.	Core data-capture function without which no balances or settlements can be computed.
FR-003	Functional	The system shall convert any expense logged in a foreign currency into the trip's base currency using the daily exchange rate in effect on the expense date.	High	Pass: Converted amount matches (expense amount x that day's published rate) within 0.01 currency units; Fail: conversion uses a stale or incorrect rate.	Multi-currency accuracy is essential for trustworthy balance calculations in international group trips.
FR-004	Functional	The system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions.	High	Pass: Group balance settles to net zero with minimal payment edges; Fail: sum of debts does not equal sum of credits.	Given (sample requirement from problem statement) — reduces the number of repayments travelers must make.
FR-005	Functional	The system shall allow a Group Traveler to mark a suggested settlement payment as 'Paid', updating that traveler's outstanding balance in real time for all trip members.	Medium	Pass: After marking a payment as Paid, both the payer's and payee's balances update within 2 seconds and are visible to all participants; Fail: balance remains unchanged or update is not reflected for other members.	Closes the settlement loop and keeps all travelers' view of outstanding balances trustworthy and current.

Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-001	Performance & Security	The settlement calculation engine must execute in under 100 ms for groups of up to 50 travelers with 500 expenses.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Given (sample requirement from problem statement) — ensures a responsive experience even for large group trips.
NFR-002	Reliability & Data Integrity	The system shall ensure that all currency conversion and settlement computations are consistent, such that recomputing the settlement graph from the same expense data always yields an identical result (deterministic, auditable calculations).	Medium	Pass: Re-running the settlement engine on an unchanged expense set produces byte-identical debt/credit values in 100/100 test runs; Fail: any variance appears between repeated runs on the same data.	Financial calculations must be deterministic and auditable so travelers trust the amounts they are asked to pay.